De La Salle University - Dasmariñas
COLLEGE OF INFORMATION AND COMPUTER STUDIES
COMPUTER SCIENCE DEPARTMENT

COURSE SYLLABUS

Second Semester, SY 2024-2025

COURSE CODE	**:**	S-CSPC428
COURSE TITLE	**:**	Networks and Communications
COURSE TYPE	**:**	Lecture
COURSE CREDIT	**:**	Three Units
LEARNING MODALITY	**:**	Traditional Blended Learning
PRE-REQUISITES	**:**	at least 3rd level
CO-REQUISITES	**:**	None
PROFESSOR	**:**	Tita Ramos Herradura
CONSULTATION HOUR	**:**	Tues.:10-12; Wed.: 10-12; Thurs.: 9-10 COS100C
CONTACT INFORMATION	**:**	trherradura@dlsud.edu.ph (046) 4811900 local 3134

VISION

A leading Catholic University that inspires excellence and drives innovation towards a just, peaceful and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research and extension programs.

COURSE DESCRIPTION:

This course deals with computer networks principles covering different layers, topologies, and models of computer networks (explore Open System Interconnection), network programming and network applications. This course also focuses on the TCP/IP protocol, network architecture (including implementation, administration, and maintenance), and infrastructure design. Students also gain "hands-on" networking experience by installing and setting up a network operating system, building a small LAN, and managing network servers. Students are also expected to do systems/network programming and make use of simulation and measurement tools to gain an appreciation of current network design and performance issues. This course utilizes the online curriculum of Networking Basics of Cisco Networking Academy Program (CNAP) @ https://www.netacad.com/.

The course will have a special provision on modality, pacing, pedagogy, assessment type, communication and feedback based on the national, regional, provincial, city, and institutional guidelines for safety amidst disaster and pandemic. Hence, this class shall be delivered in both online and offline/module-based approach. This course is also compliant with Outcomes-Based Education (OBE) and 21st Century Learning Design (21CLD) frameworks.

COURSE LEARNING OUTCOMES:

By the end of this course, students can:
CLO1. explain important concepts in network communication;
CLO2. explain network types, components, and connections;
CLO3. explain how communication occurs on Ethernet networks;
CLO4. explain the features of an IP address;
CLO5. explain how routers connect networks together;
CLO6. explain the function of transport and application layer services;
CLO7. use various tools to test and troubleshoot network connectivity;
CLO8. configure mobile devices for wireless access; and
CLO9. create a fully connected LAN

LEARNING PLAN:
Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
CLO1. Explain important concepts in network communication.	TLO1. Explain the concept of network TLO2. Explain network transmission speed and capacity. .
CLO2. Explain network types, components, and connections.	TLO3. Explain the roles of network infrastructure devices. TLO4. Describe ISP connectivity options.
CLO3. Explain how communication occurs on Ethernet networks	TLO5. Describe network communication protocols and standards. TLO6. Explain the process of encapsulation and Ethernet framing.
CLO4. Explain the features of IP address	TLO7. Explain how IPv4 and IPV6 addresses are used in network communication and segmentation. TLO8. Configure a DHCP server.
CLO8. Configure mobile devices for wireless access.	TLO9. Describe the different types of networks used by cell phones and mobile devices. TLO10. Describe the components required to build a home network. TLO11. Describe wired and wireless network technologies.

Module No and Learning Outcome(s)	**Week Date/s**	**Teaching-Learning Lessons and Activities / Assessment Strategies**	**Technology Enabler**	**Onsite**	**Asynchronous**	**Allotted Hours**

Module 1 TLO 1 TLO 2 TLO 3 TLO 4 CLO 1 CLO 2	WEEK 1 (1/20-1/25)	• Module 1: Build a Small Network • Gospel, Reflection and Prayer • Presentation of course introduction • Course requirement details • Introduction - Communication in a Connected World • Class Participation (group discussion of concept of network communication)	Schoolbook MSTeams	/		3
	WEEK 2 (1/27 - 2/1)	• Module 1: Build a Small Network • Gospel, Reflection and Prayer • Build a Small Network • Network Components, Types, and Connections. • Class Participation (identity the network components and types)			/	3
	WEEK 3 (2/3-2/8)	• Gospel, Reflection and Prayer • Enabling assessment (explain the concept of network communication and build a simple network)		/		3
Module 2 TLO 9 TLO 10 TLO 11 CLO 8	WEEK 4 (2/10-2/15)	• Module 2: Network Access • Gospel, Reflection and Prayer • Network Access • Communication Principles, Protocols and Standards • Class Participation (group discussion of the importance of standards and protocols in network communications)	Schoolbook MSTeams		/	3
	WEEK 5 (2/17-2/22)	• Module 2: Network Access • Gospel, Reflection and Prayer • Network Media • Enabling assessment (explain the different network media)		/		3
Module 3 TLO 7 TLO 8 CLO 4	WEEK 6 (2/24-3/1)	• Module 3: The Internet Protocol • Gospel, Reflection and Prayer • The Internet Protocol • Class Participation (explain the features of IP address)			/	3

	WEEK 7 (3/3-3/8)	• Module 3: The Internet Protocol • Gospel, Reflection and Prayer • IPV4 and IPV6 addressing • Class Participation (explain the principles of IPv4 and IPv6 address management)	Schoolbook MSTeams	/		3
	WEEK 8 (3/10-3/15)	• Module 3: The Internet Protocol • Gospel, Reflection and Prayer • Dynamic addressing with DHCP • Enabling assessment (configure a DHCP server)			/	3
	9 (3/17-3/22)	MIDTERM EXAMINATION (Summative assessment – covers Module 1-3)				3
			TOTAL	***12.0***	***12.0***	***27.0***

Final Period

Course Learning Outcomes	Topic Learning Outcomes
CLO5. Explain how routers connect networks together.	TLO12. Describe network boundaries. TLO13. Explain the purpose of Network Address Translation in small networks.
CLO6. Explain the function of transport and application layers.	TLO14. Explain how clients access internet services. TLO15. Describe common network applications.
CLO7. Use various tools to test and troubleshoot network connectivity.	TLO16. Troubleshoot using network utilities.
CLO9. Create a fully connected LAN.	TLO17. Explain the need for routing. TLO18. Explain how routers use tables.

Module No and Learning Outcome(s)	**Week Date/s**	**Teaching-Learning Lessons and Activities / Assessment Strategies**	**Technology Enabler**	**Onsite**	**Asynchronous**	**Allotted Hours**

Module 4 TLO 12 TLO 13 TLO 17 TLO 18 CLO 5 CLO 9	SUSTAINABILITY WEEK 10 (3/24-3/29)	• Module 4: Communication Between Networks • Gospel, Reflection and Prayer • Communication Between Networks • Gateways to Other Networks • Class Participation (Explain how routers connect networks together)	Schoolbook MSTeams		/	
	WEEK 11 (3/31-4/5)	• Module 4: Communication Between Networks • Gospel, Reflection and Prayer • Address Resolution • Class Participation (Explain how ARP enables communication on a network)			/	
	WEEK 12 (4/7-4/12)	• Module 4: Communication Between Networks • Gospel, Reflection and Prayer • Routing Between Networks • Class Participation (Create a fully connected LAN)	Schoolbook MSTeams	/		3
	WEEK 13 (4/14-4/19)	• Independent learning days/Holy week			/	3
Module 5 TLO 14 TLO 15 TLO 16 CLO 6 CLO 7	RESEARCH WEEK 14 (4/21-4/26)	• Module 5: Protocols for Specific Tasks • Gospel, Reflection and Prayer • Protocols for Specific Tasks • Transport Layer • Enabling assessment (Explain how clients access internet services)	Schoolbook MSTeams		/	3
	WEEK 15 (4/28-5/3)	• Module 5: Protocols for Specific Tasks • Gospel, Reflection and Prayer • Application Layer Services and Troubleshooting • Class Participation (Explain the function of common application layer services; Use various tools to test and troubleshoot network connectivity)	Schoolbook MSTeams	/		3
	WEEK 16 (5/5-5/10)	FINAL EXAMINATION (Summative assessment)		/		3
			TOTAL	***9.0***	***12.0***	***21.0***

GRADING SYSTEM:

Midterm	%	Final	%
Enabling Assessments	50%	Enabling Assessments	50%
Summative Assessment	30%	Summative Assessment	30%
Class Participation	20%	Class Participation	20%
	100%		100%

FINAL GRADE = $\frac{\text{Midterm Grade + Final Grade}}{2}$

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Traditional Blended Learning Mode. There will be 50% onsite and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
    a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered face-to-face while the remaining is asynchronous.
    b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
        i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
        ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through the Schoolbook messaging feature.
        iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
    c. No use of foul, curse, and other bad words, whether written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
    a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for the student's grade.
    b. The onsite meeting will only be held at the schedule indicated in the registration form.
    c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
    a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.

c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
d. Giving passwords to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 accounts other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
   a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
   b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
   c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless special permission is sought. Cellular phones cannot also be used as calculators during examination.
   d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

REFERENCES:

Online References

Call number or e-provider	**Reference Material**
www.netacad.com	Cisco Networking Academy, Retrieved from https://www.netacad.com/
	Adele Kuzmiakova. 2021. Computer Networks and Communications. Arcler Press. https://research.ebsco.com/c/lleopn/search/details/ehdv6ex7pf?limiters=FT%3AY&q=networks+and+communications
	Haas, H., Islim, M. S., Chen, C., & Abumarshoud, H. 2021. An Introduction to Optical Wireless Mobile Communications. Artech House. https://research.ebsco.com/c/lleopn/search/details/ehwa2gyvuf?limiters=FT%3AY%2CDT1%3A2020-01-18%2F2025-01-18&q=networks%20and%20communications
Journal	Tang, J., Shao, S., Guo, S., Wang, Y., & Wu, S. (2024). A Collaborative Allocation Algorithm of Communicating, Caching and Computing Resources in Local Power Wireless Communication Network. Information (2078-2489), 15(6), 309. https://doi.org/10.3390/info15060309

Prepared: **Computer Science Department**
Second Semester 2024-2025

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